

Image Data Analysis & Machine Learning in *Orange*

A promotional banner for Orange Data Mining software. The background is a solid purple color. On the left, the text "Data Mining Fruitful and Fun" is written in a large, white, sans-serif font. Below this, in a smaller white font, is the text "Open source machine learning and data visualization." At the bottom left, there is an orange button with the text "Download Orange 3.40.0" in white. On the right side of the banner, there is a cartoon illustration of an orange character with large black-rimmed glasses, a green leaf on its head, and a wide smile. The character is holding a string of several colorful balloons (blue, pink, orange) that are floating upwards. Each balloon has a different icon representing data science concepts: a lightbulb, a bar chart, a tree diagram, a scatter plot, a line graph, and a network diagram. The overall style is playful and educational.

**Data Mining
Fruitful and Fun**

Open source machine learning and data
visualization.

Download Orange 3.40.0

Find our materials here→
<http://tiny.cc/cuaiorange>

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How many of you can do programming?

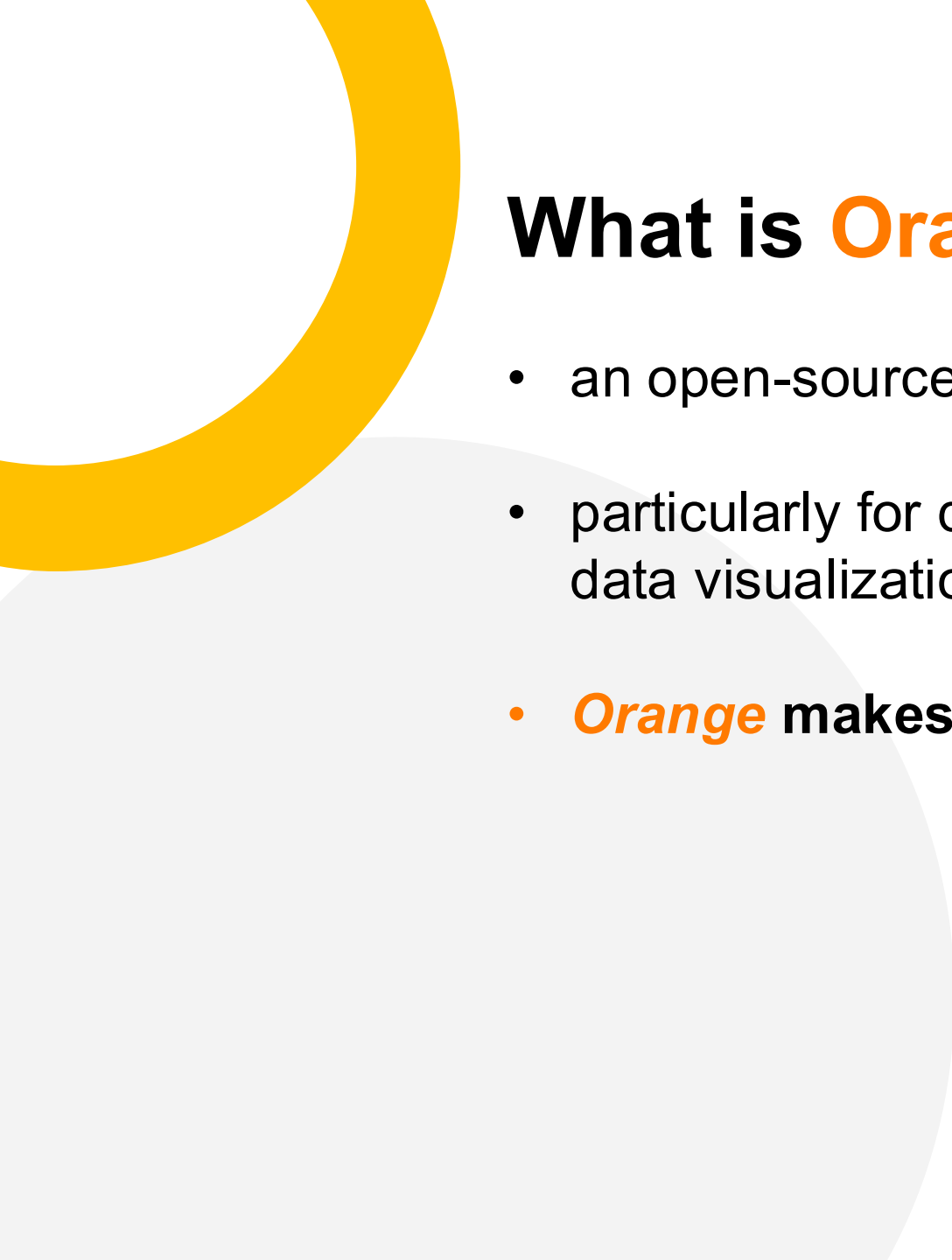
**Hands
up**



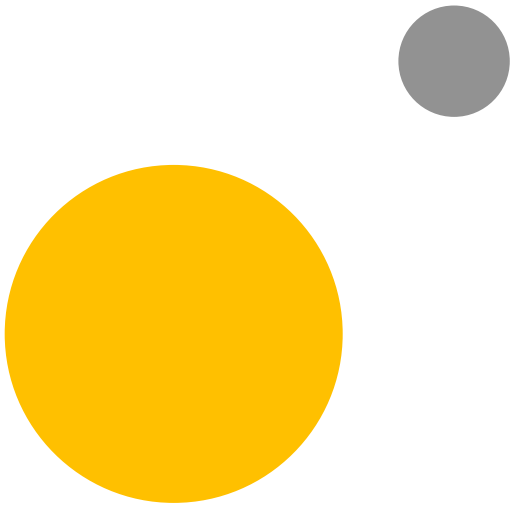
How many of you enjoy programming?

**Hands
up**





What is **Orange**?

- an open-source data visualization and analysis tool.
 - particularly for data mining, machine learning, and data visualization.
 - **Orange** makes everything simple! :)
- 

Video tutorials:

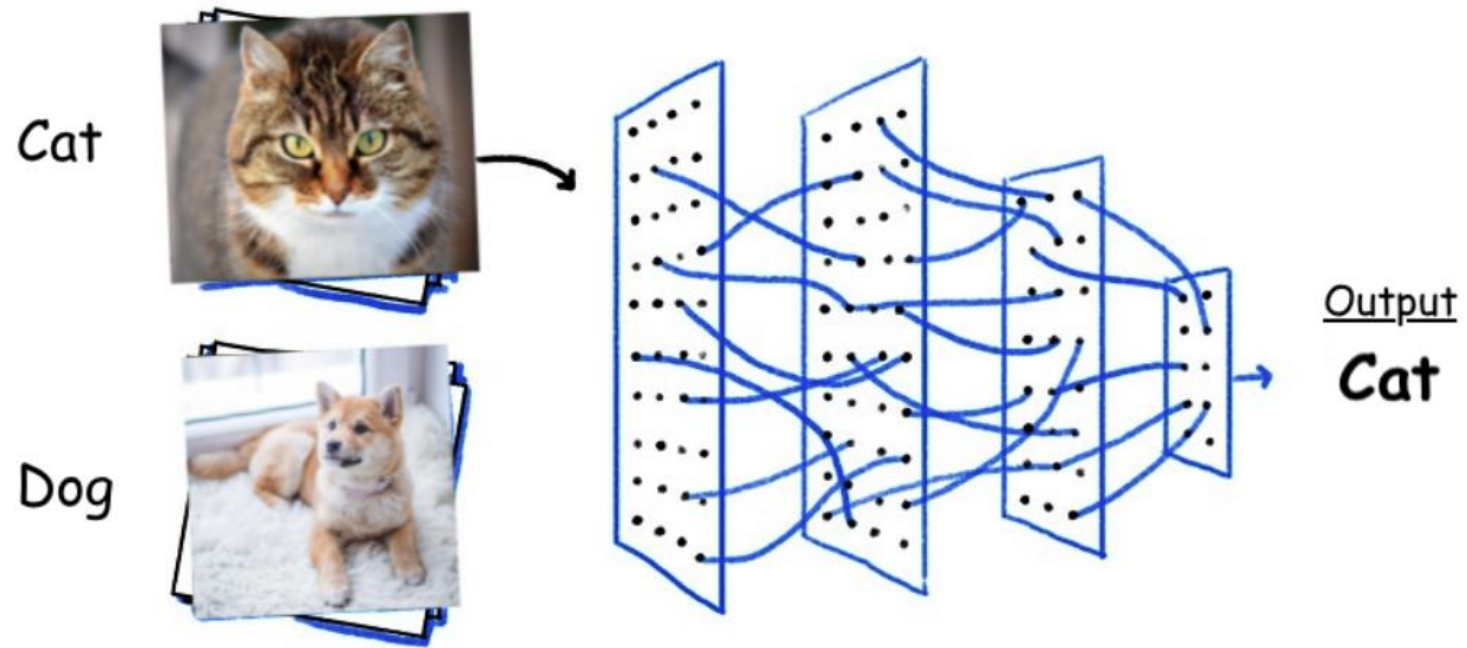
<https://www.youtube.com/watch?v=lb-x36xqJ-E&list=PLmNPvQr9Tf-ZSDLwOzxpY-HrE0yv-8Fy&index=2>

You can start **Orange** from here!

The official website offers very detailed tutorials on creating the data visualizations you want step by step.

1	 Getting Started with Orange Welcome to Orange 2:16	Getting Started with Orange 01: Welcome to Orange Orange Data Mining • 100万次观看 • 8年前	⋮
2	 Getting Started with Orange Data Workflows 2:35	Getting Started with Orange 02: Data Workflows Orange Data Mining • 40万次观看 • 8年前	⋮
3	 Getting Started with Orange Widgets and Channels 3:13	Getting Started with Orange 03: Widgets and Channels Orange Data Mining • 25万次观看 • 8年前	⋮
4	 Getting Started with Orange Loading Your Data 4:15	Getting Started with Orange 04: Loading Your Data Orange Data Mining • 26万次观看 • 8年前	⋮
5	 Getting Started with Orange Hierarchical Clustering 4:03	Getting Started With Orange 05: Hierarchical Clustering Orange Data Mining • 22万次观看 • 8年前	⋮
		Getting Started with Orange 06: Making Predictions	

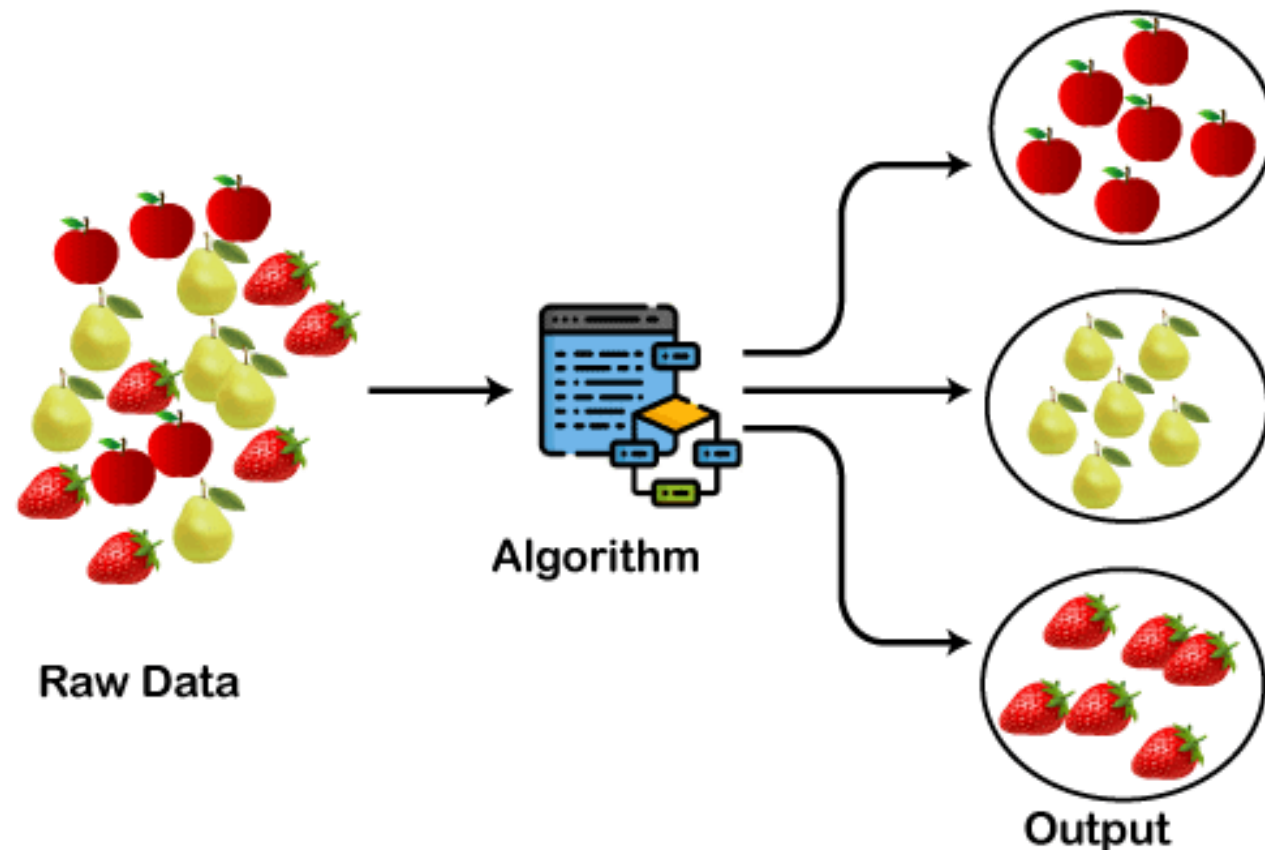
Image Data Analysis



What can Orange do with images?

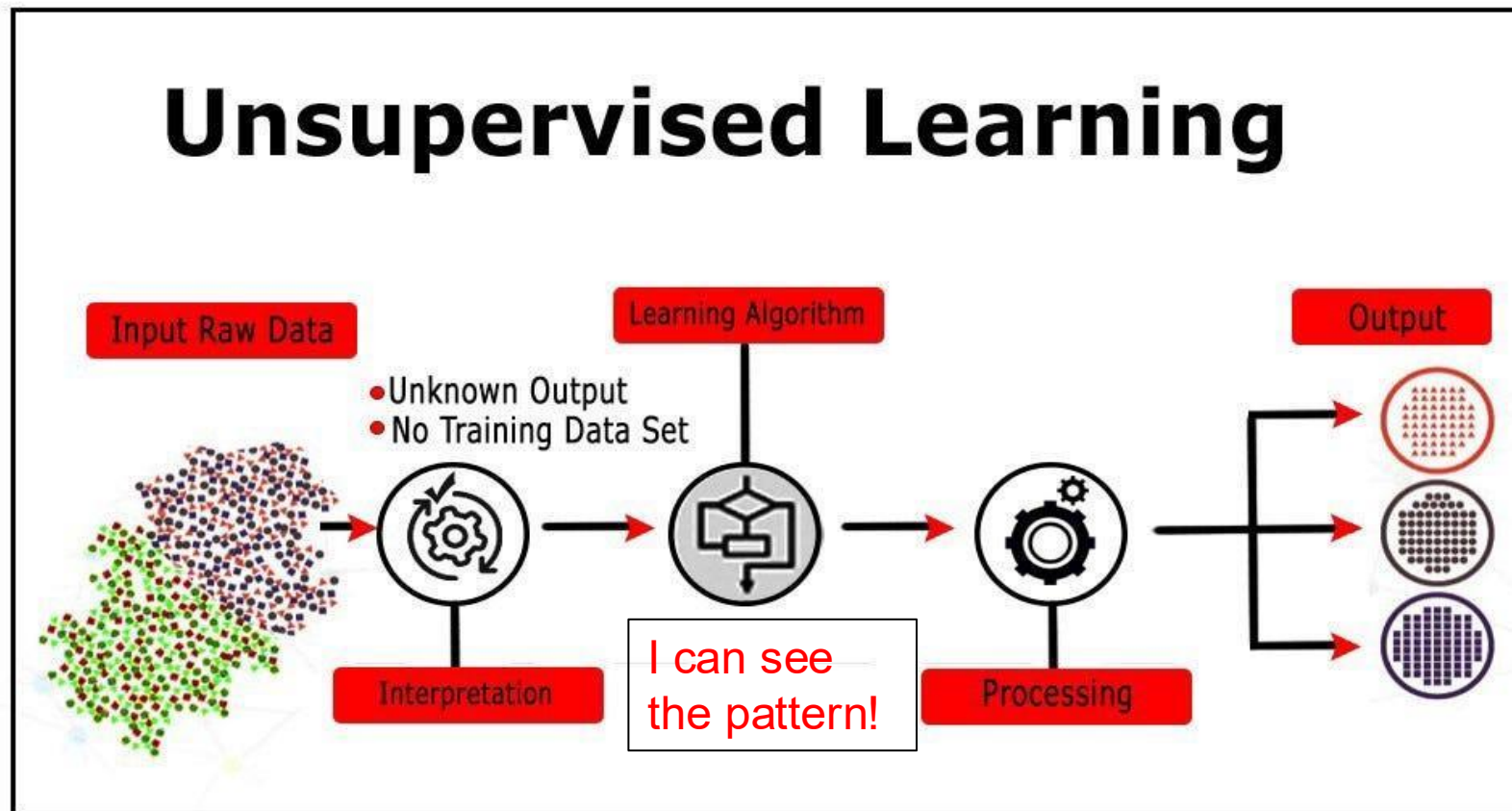
Hierarchical Clustering

- An **unsupervised machine learning** technique used to group similar data points together.
- To find natural, hidden patterns or structures in data without any pre-existing labels.



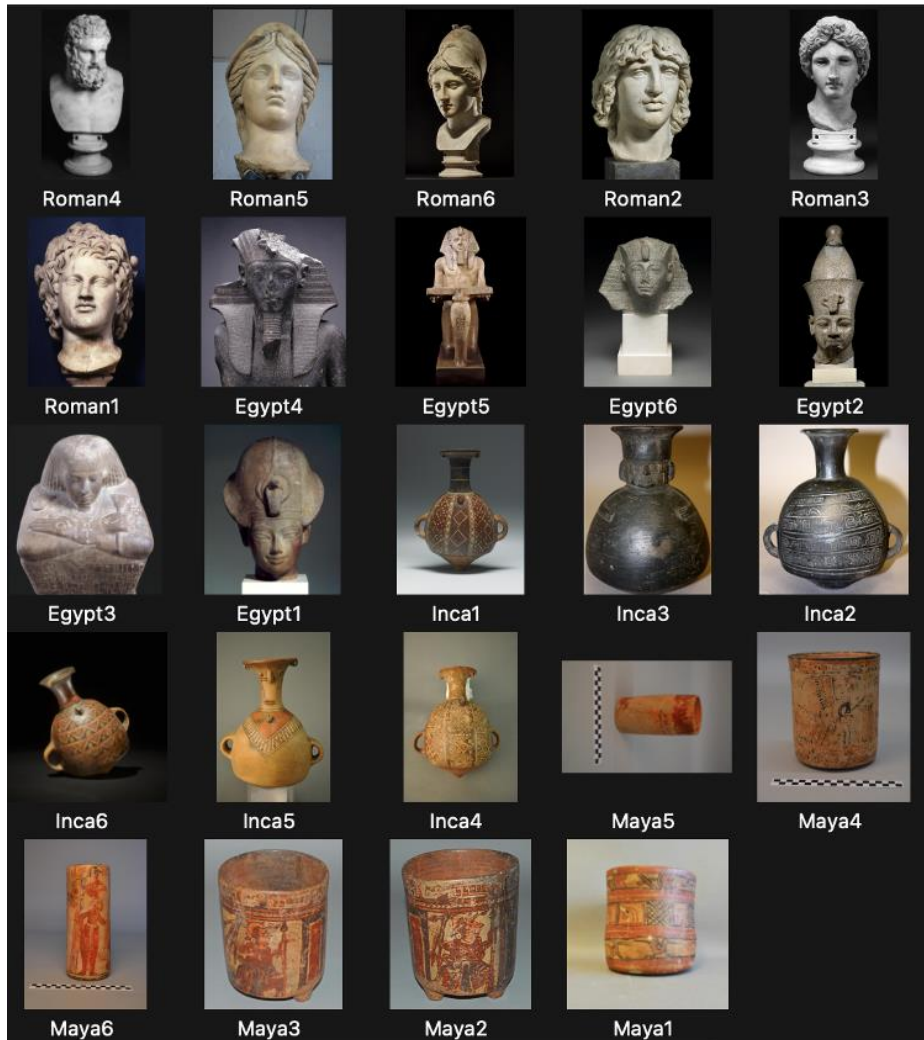
Why is it “machine learning”?

- It's a data-driven approach; the machine “learns” the grouping rules from the data itself.
- The algorithm improves its grouping with each iteration to find the optimal structure.



Today's Goal

Using hierarchical clustering to perform image analysis by Orange



Install - operate - interpret – self-practice –
brainstorming

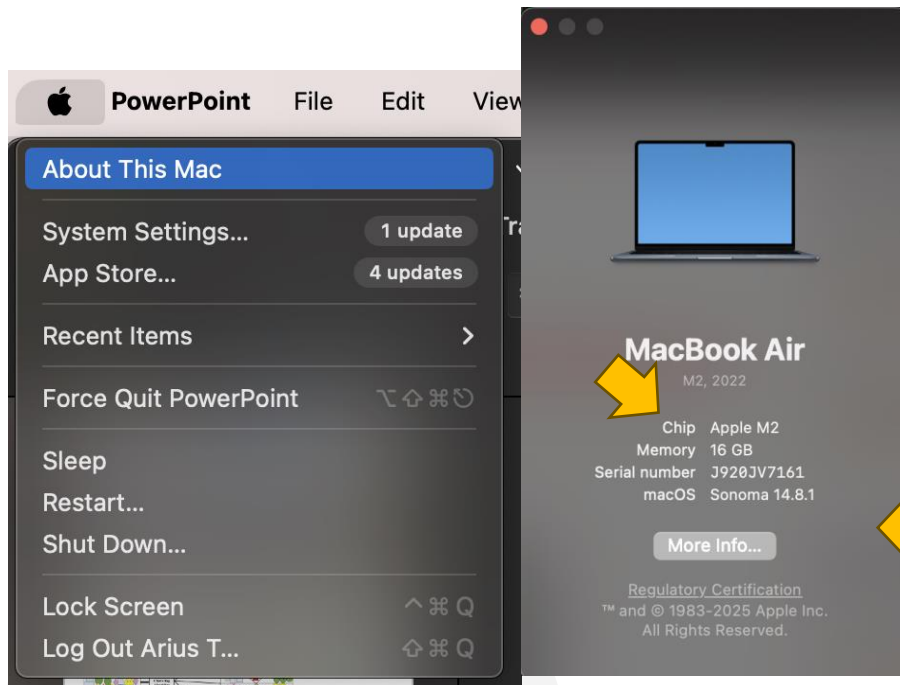
Steps: 1. Install Orange

Windows users: <https://download.biolab.si/download/files/>

[Orange3-3.38.0-Python3.10.11-x86_64.dmg](#)
[Orange3-3.38.0-Python3.11.8-arm64.dmg](#)
[Orange3-3.38.0-Python311-win_amd64.exe](#)
[Orange3-3.38.0.zip](#)
[Orange3-3.38.1-Miniforge-x86_64.exe](#)
[Orange3-3.38.1-Python3.10.11-x86_64.dmg](#)
[Orange3-3.38.1-Python3.11.8-arm64.dmg](#)
[Orange3-3.38.1-Python311-win_amd64.exe](#)
[Orange3-3.38.1-problematic-link.zip](#)
[Orange3-3.38.1.zip](#)
[Orange3-3.39.0-Python3.10.11-x86_64.dmg](#)

Download Orange3-3.38.1-Python311-win_amd64.exe

Apple users: <https://orangedatamining.com/download/>



macOS

Orange for Apple silicon

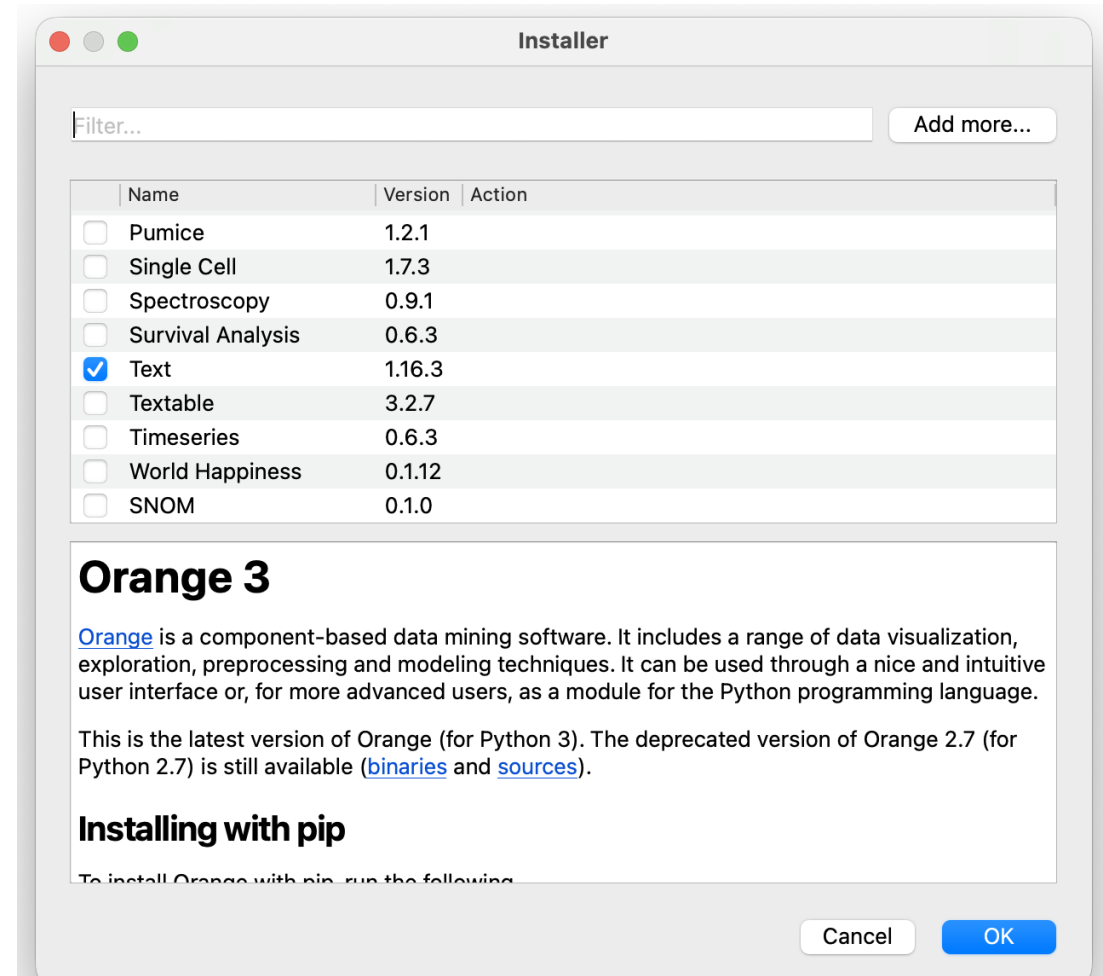
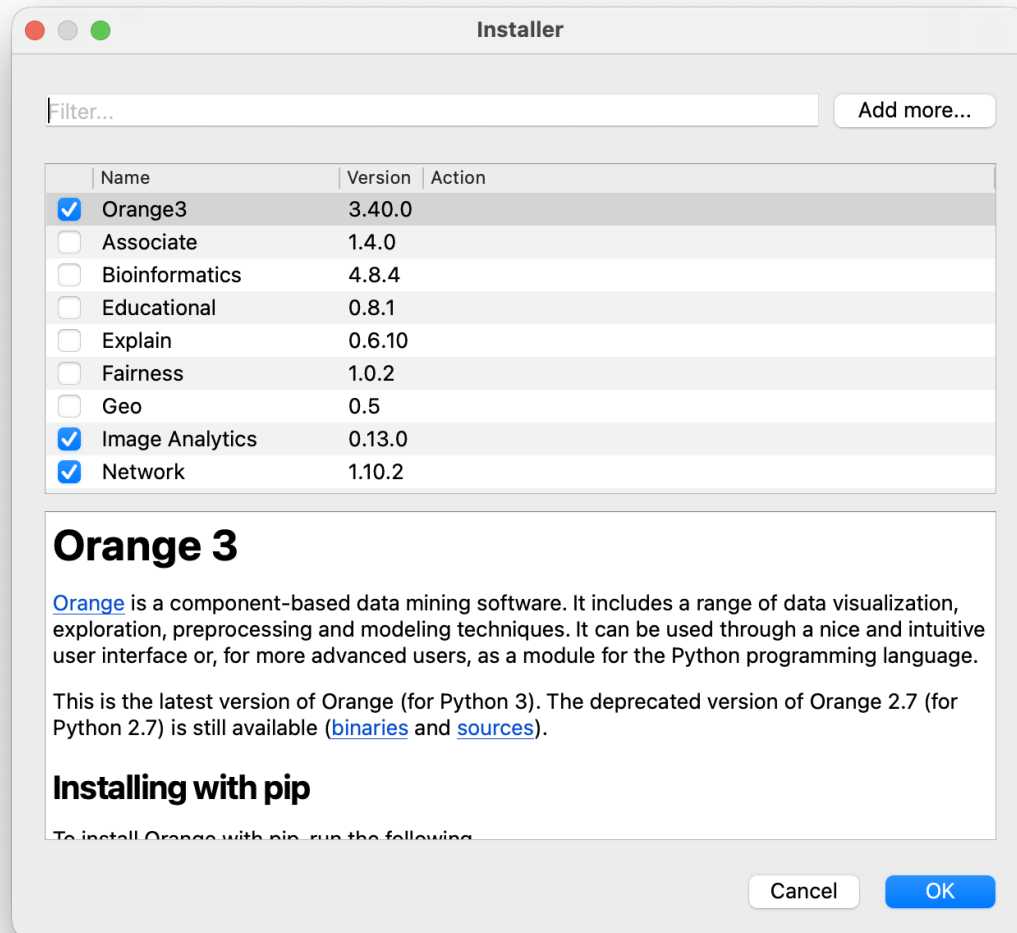
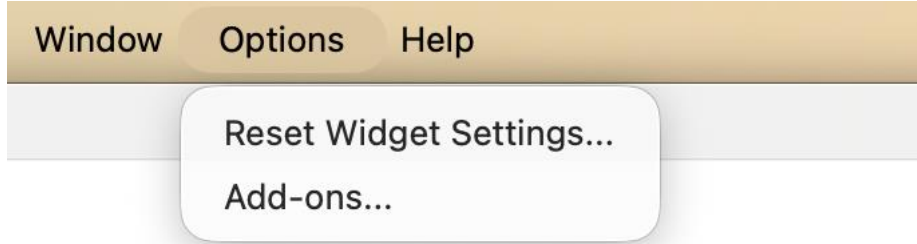
↓ [Orange3-3.40.0-Python3.12.10-arm64.dmg](#)

Orange for Intel

↓ [Orange3-3.40.0-Python3.12.10-x86_64.dmg](#)

Not sure which installer to select? Click the Apple logo in the top-left corner of your screen, select About This Mac, and check the Chip or Processor field. If you see Apple, select the Orange for Apple Silicon installer. If you see Intel, select the Orange for Intel.

2. Open Orange- options - 'add-ons'



3. Download two image folders here:

<https://tinyurl.com/35z76rse>



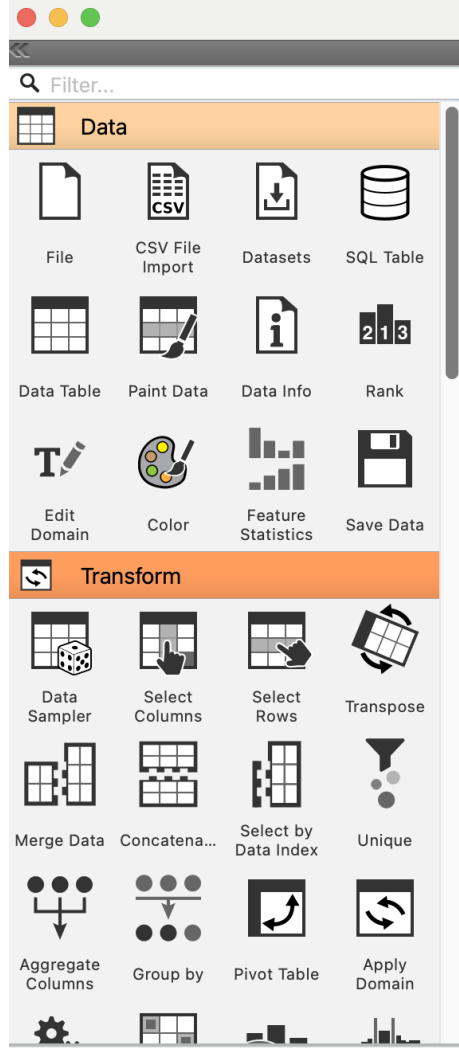
Islamic Coins VS Chinese Coins



Statues VS Vases

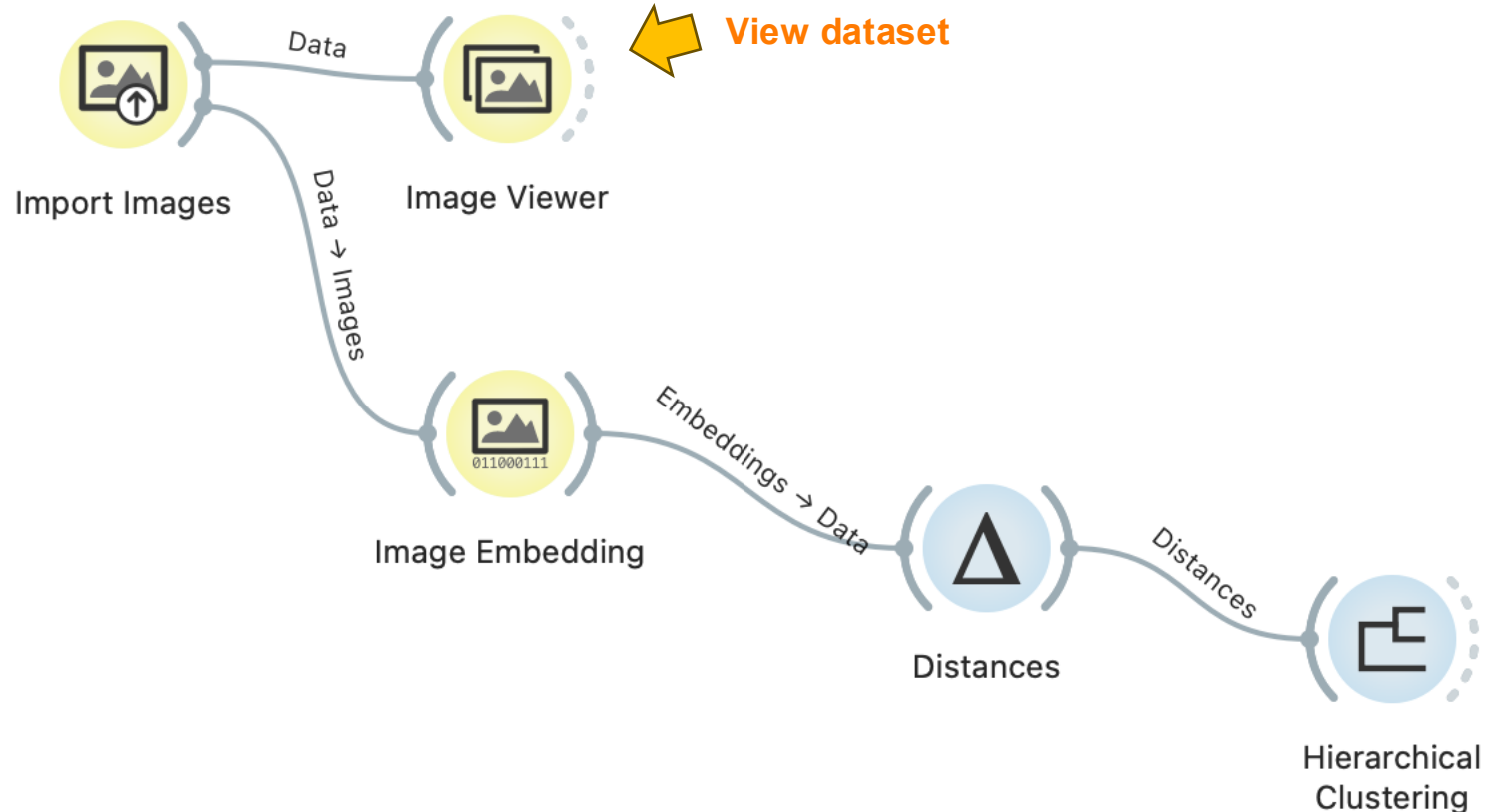
4. Use the folder “Islamic Coins VS Chinese Coins”

Follow the procedure map below:



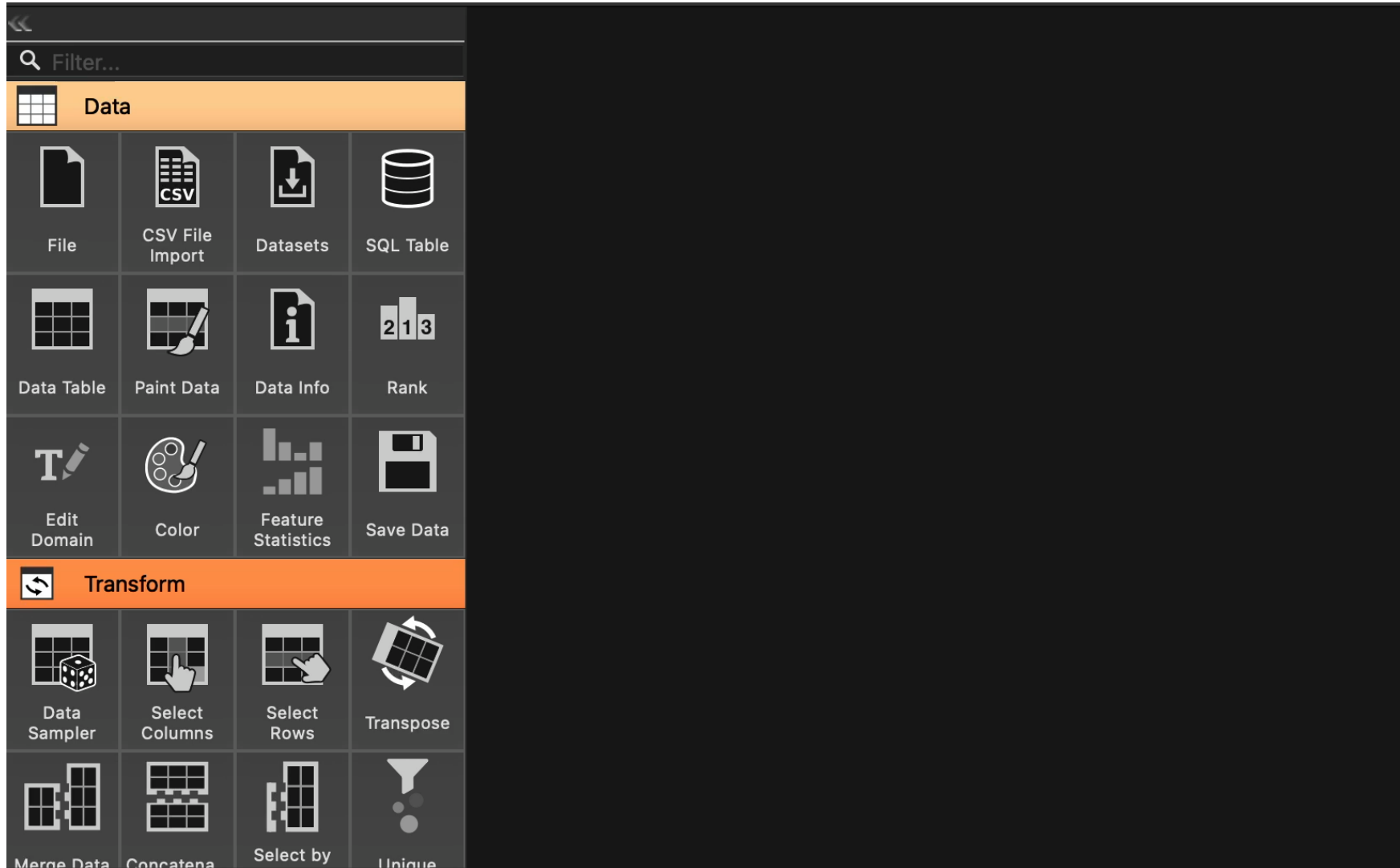
Select a widget to show its description.

See [workflow examples](#), [YouTube tutorials](#), or open the [welcome screen](#).



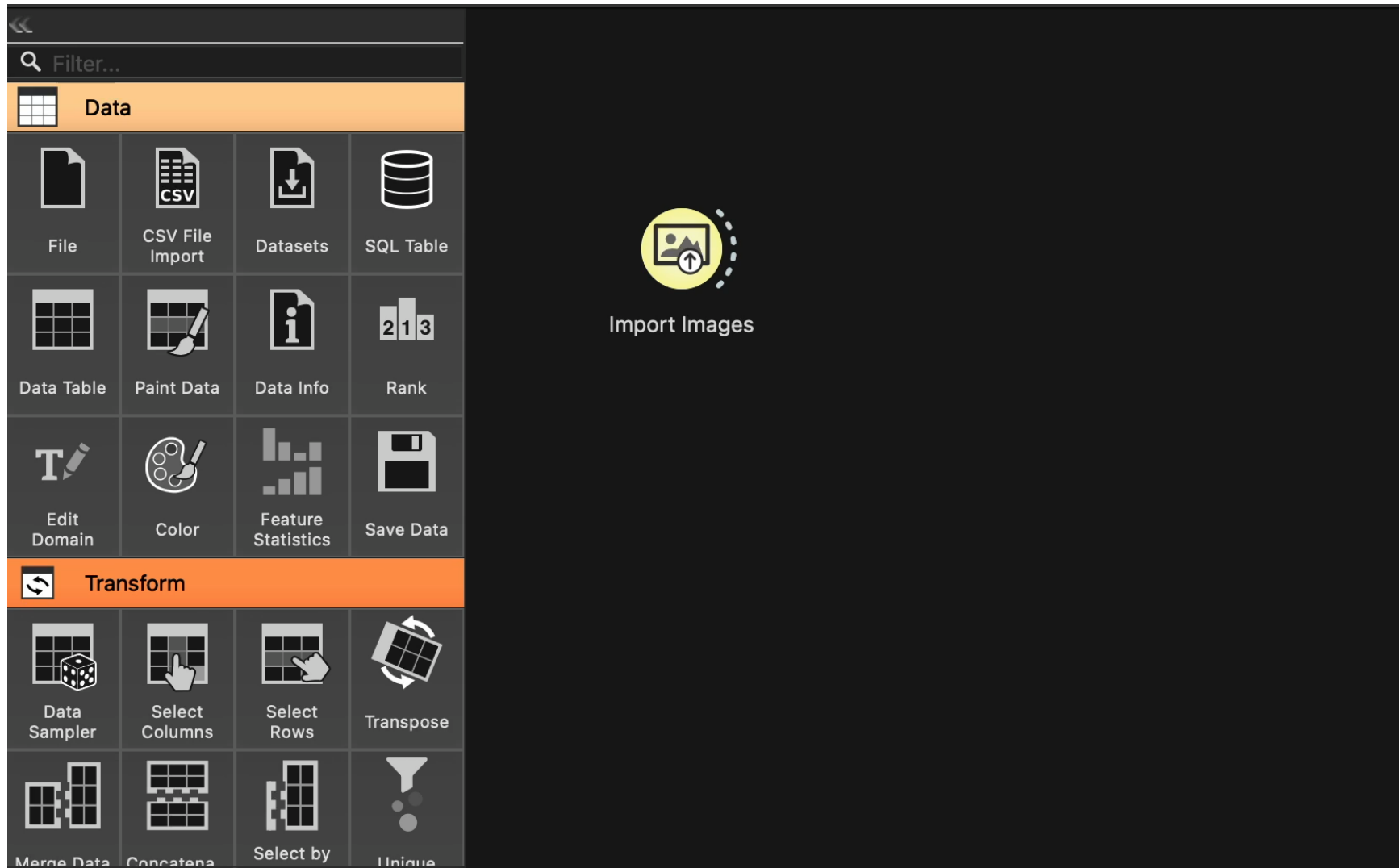
Step	Purpose	Example (Islamic & Chinese Coins)
Import images	Establish the raw dataset of research objects	Import a mixed collection of Islamic and Chinese coin images to build the corpus for analysis
Image embedding	Transform unstructured visual content into structured numerical features (comparable "feature vectors")	Let the computer automatically extract visual characteristics of each coin (e.g., shape, script patterns, texture), generating comparable feature indicators
Distances	Quantify similarity/dissimilarity between samples	Calculate visual distances between all pairs of coins; smaller values indicate higher visual similarity, forming a similarity matrix
Hierarchical Clustering	Discover underlying structures and categorical patterns in the data through unsupervised learning	Allow coins to automatically group based on similarity, generating a “family tree” that reveals potential classification systems and exceptional samples

To create initial module:



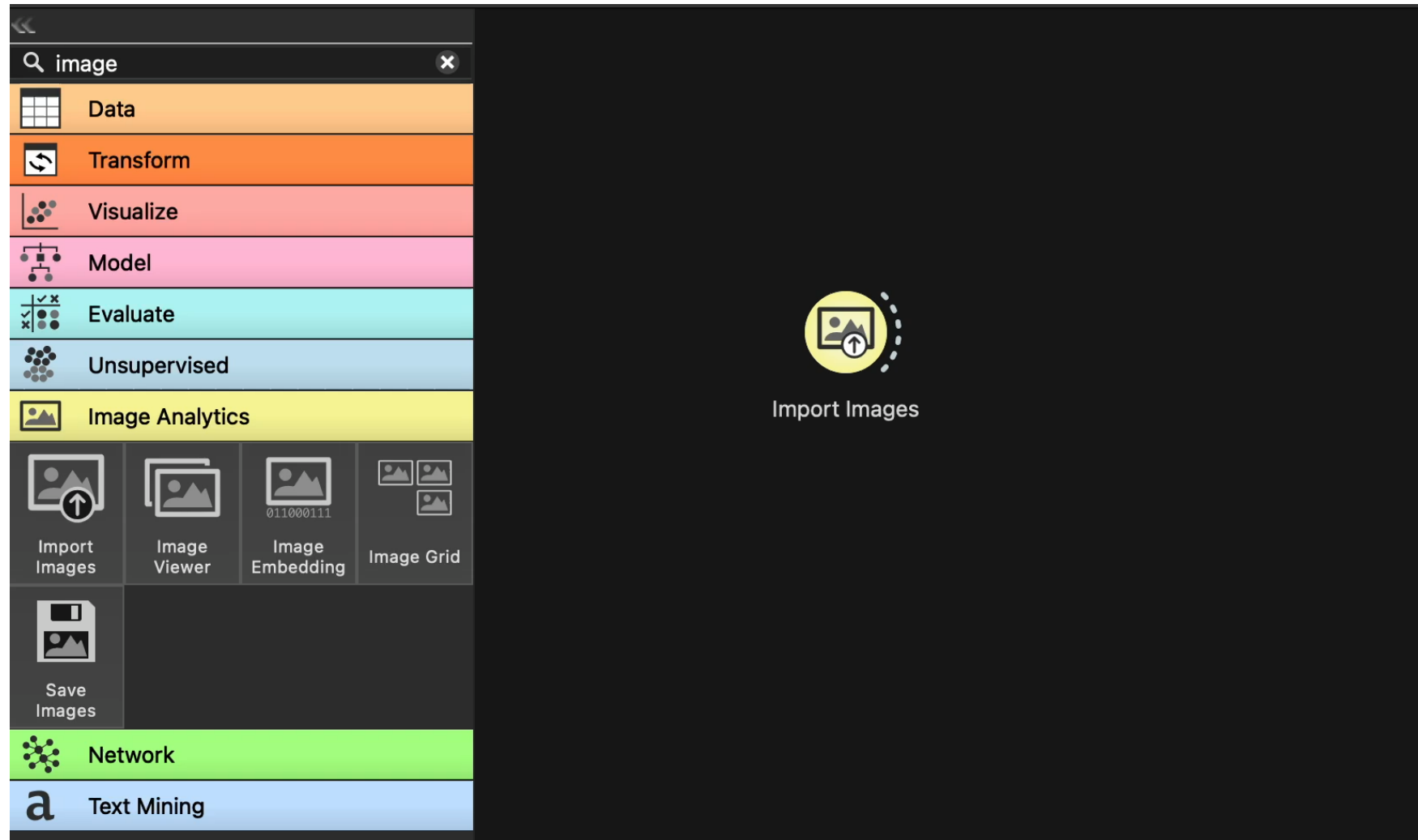
To create next module(s):

Method 1

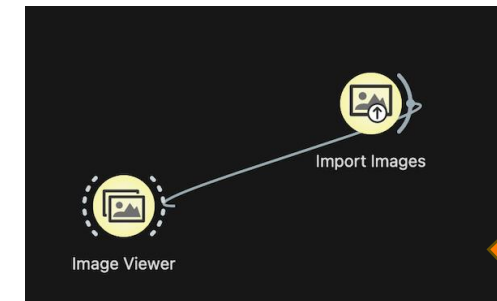


To create next module(s):

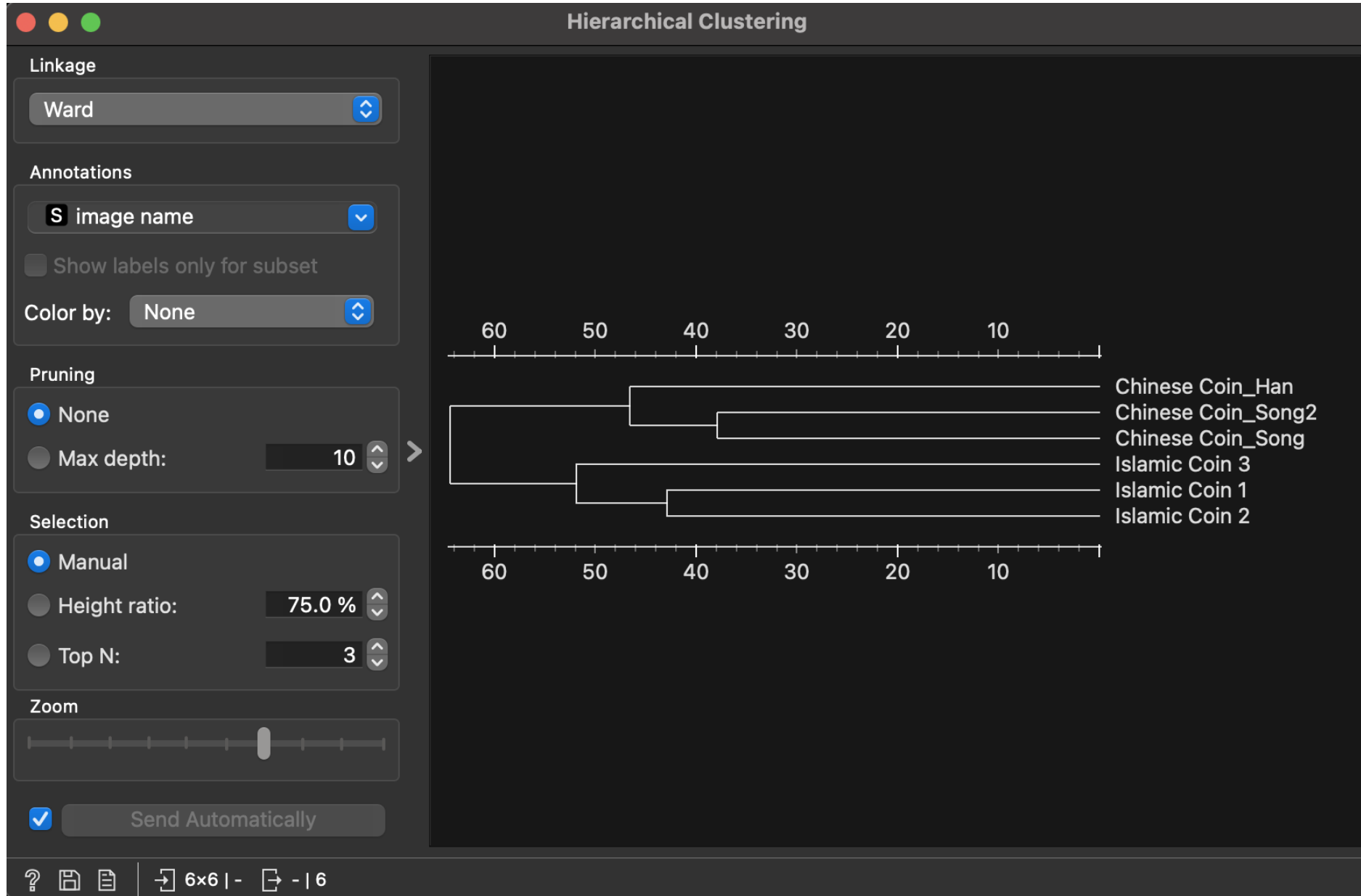
Method 2



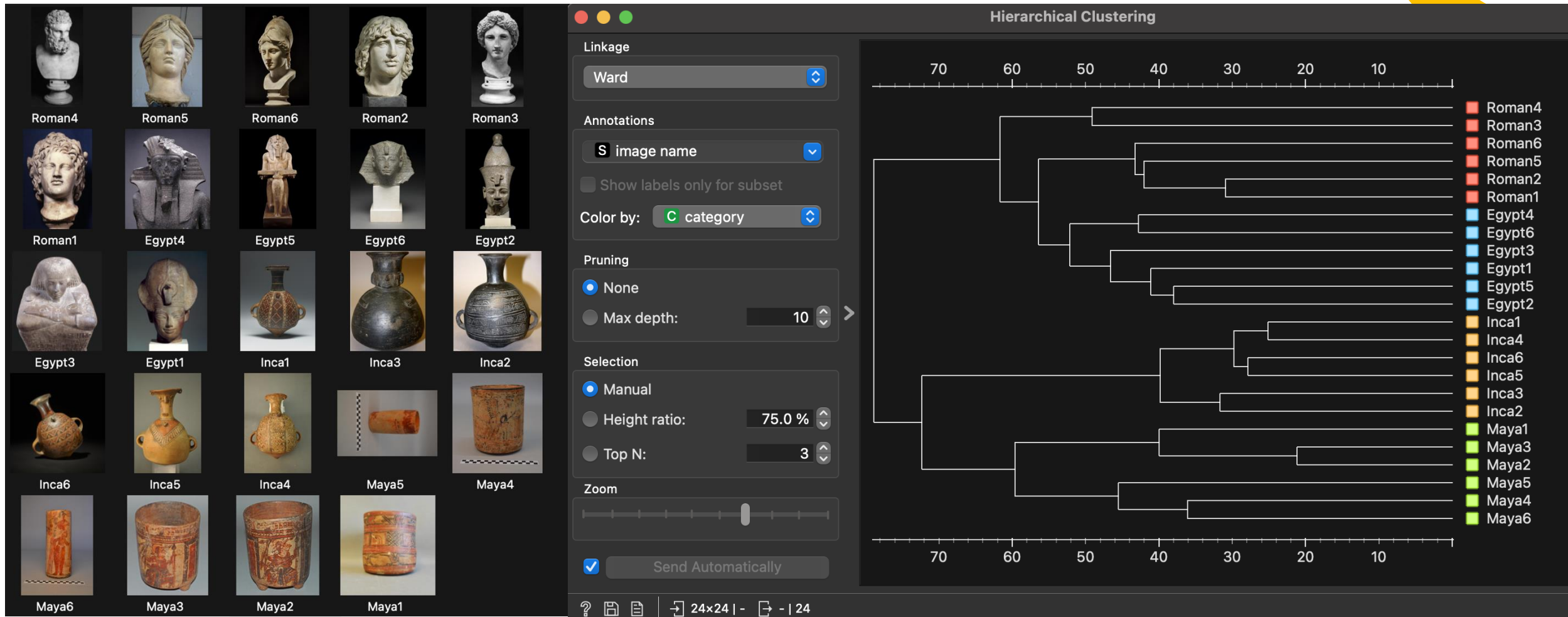
***Make sure the next module is on the right side and connected correctly!**



5. Double click “Hierarchical Clustering”



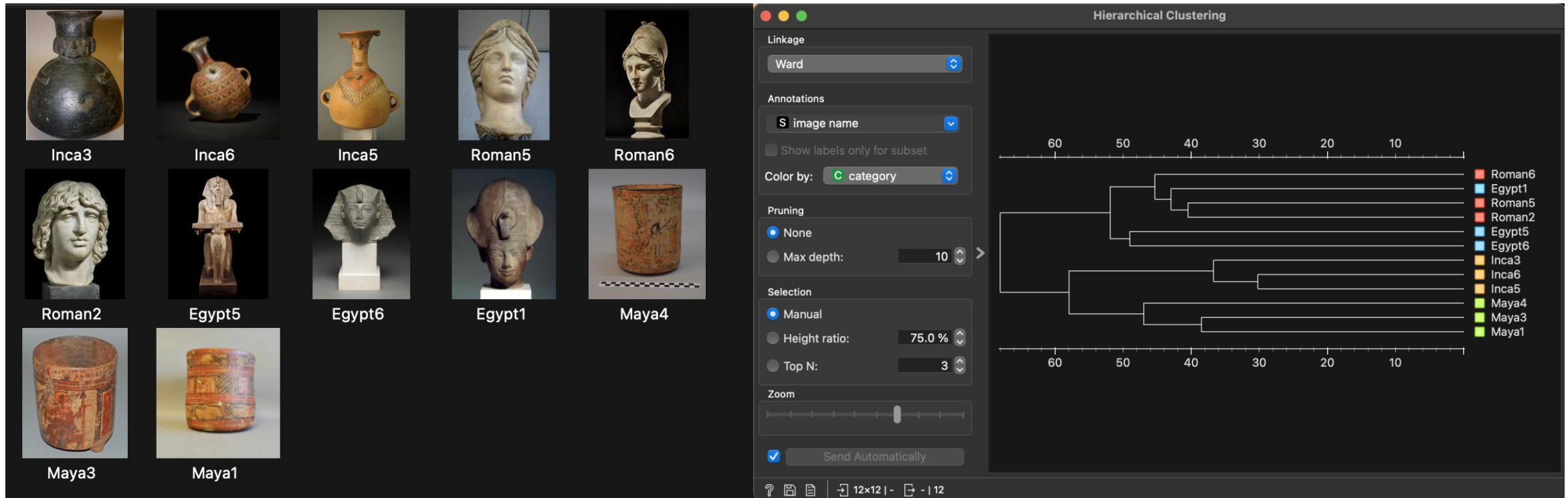
6. Use the folder "Statues VS Vases" to self-practice



Questions:

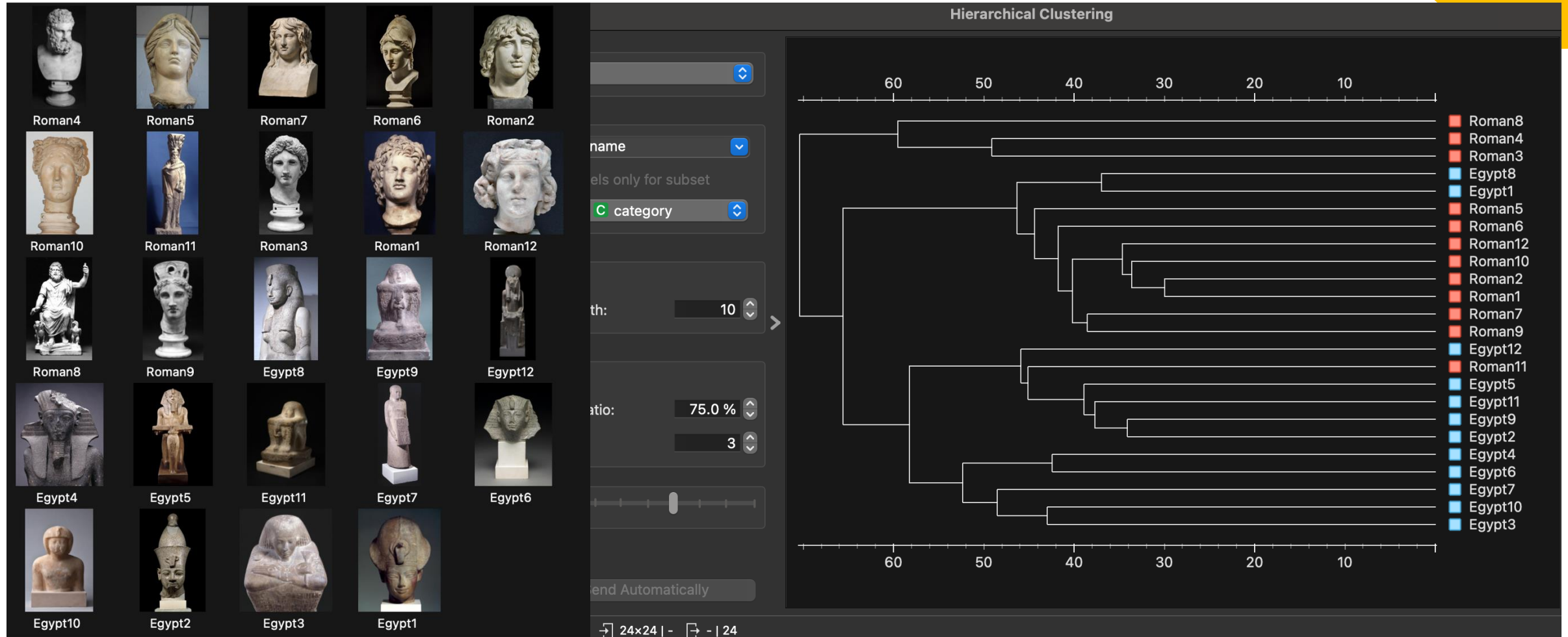
1. How can we replace existing dataset (Coins) with the new dataset?
2. How can we “create” the colour blocks on the clustering network?

Advanced Question- case 1:



Why is the result inaccurate (i.e. Egypt and Roman statues sorted into one category)?

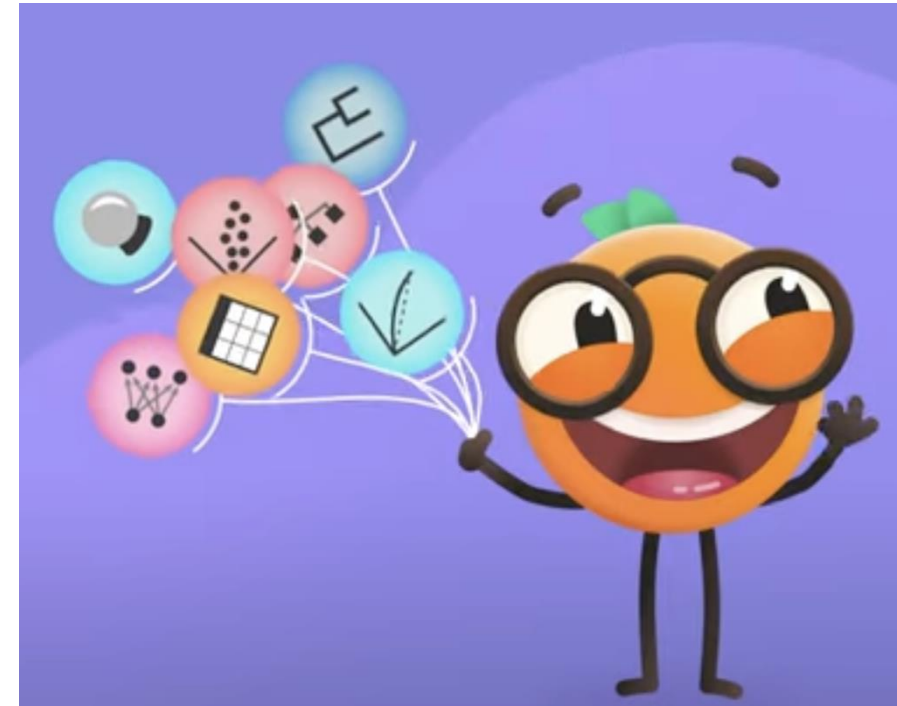
Advanced Question- case 2:

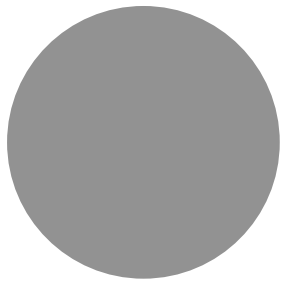


Why is the result inaccurate (i.e. Egypt and Roman statues sorted into one category)?

There are even more exciting features waiting for you to explore!

Get started now and embark on your exploration journey!





Thank you!

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